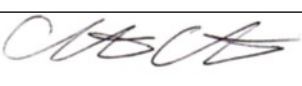


Guideline Title: Pediatric Reactive Airway Disease

Guideline Number: GUID-3203-PED014

Issue date: 09/09/2014	Replaces Dept. Policy:
Revision dates: 4/30/24	Developed by: Air Care Team
Approved by: Dr. Christopher Hunter, MD Air Care Team Medical Director	Approved by:
Signature: 	Signature:
Department Numbers: 3203	

- I. **PURPOSE:** To relieve bronchospasm and decrease airway edema and mucous production in pediatric patients with reactive airway disease.
- II. **PATHOPHYSIOLOGY:** Reactive airway disease represents the constellation of signs and symptoms including bronchospasm, airway inflammation, and mucous production. Severe exacerbations are those that produce an FEV1 or PEFr < 50% of the patient's predicted best. The cause of the process may be intrinsic (infection, cold air, exercise) or extrinsic (seasonal/allergic component).
- III. **DEFINITIONS:**
- IV. **DEPARTMENT GUIDELINE:**
 - A. Assess airway, breathing, and circulation as per GUID3203-PED001 – *Pediatric General Management*.
 - B. Monitor ETCO2 if intubated. Optimize oxygen content – Keep O2 sat 95%.
 - C. Assess for airway patency, anxiety/agitation, heart rate, pulse quality, skin perfusion, respiratory distress (retractions, grunting, and use of accessory muscles), and level of consciousness.
 - D. Gather history of asthma, including prior hospital admission or intubations
 - E. If endotracheal intubation is needed, consider using **Ketamine 1-4.5 mg/kg (start at 2mg/kg) IV/IO** as the induction agent per GUID3203-G008 – *Rapid Sequence Intubation*
 - F. Use caution to provide an adequate expiratory time with mechanical ventilation
 - G. Treat with inhalation therapy per GUID3203-PR012 – *Inhalation Therapy*
 - i. Nebulizer: **Pediatric > 12 years old: Albuterol 2.5 mg/3ml q 20 min** may repeat x3. If needed, continuous nebulizer dose: 10-15 mg/h
 - ii. Nebulizer: **Pediatric < 12 years old: Albuterol 0.15-0.3 mg/kg q 20 min** may repeat x3. If needed, continuous nebulizer dose: 0.5 mg/kg/h
 - iii. May add **Ipratropium bromide 0.02% 0.5 mg (2.5 ml)** to the initial albuterol nebulized x 1 dose.
 - H. Consider **methylprednisolone 1-2 mg/kg IV (max dose 60 mg)** if the patient has not had oral or parenteral bolus steroids at the sending institution. Treat patients given PO steroids who vomit within 30 minutes of medication administration.
 - I. For severe bronchospasm consider **magnesium sulfate 50 mg/kg IV** over 15-30 minutes, max dose 2 Grams.
 - J. Beta-agonists are not routinely utilized. If needed, consider giving either:

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- i. **Terbutaline 0.25 mg (> 12 yrs old) or 0.005-0.01 mg/kg (< 12 years old) SC.** May repeat in 15 minutes. Max dose 0.5mg in 4 hours
 - ii. **Epinephrine (1:1,000) 0.01 mg/kg IM** to 0.5mg max single dose, repeated prn every 20 minutes to 3 doses total. Preferred IM site is anterior thigh.
 - K. Trial Bi-pap if at sending facility: Set pressure support at 10 and PEEP at 5 if patient is alert with intact airway reflexes, not vomiting, and a mask fits appropriately.
- I. **DOCUMENTATION:**
 - A. EMS Charts
- II. **REFERENCES:**
 - A. Pediatric Advanced Life Support Manual, 2020. American Heart Association.